

The upstream placebo: reservoir siting, not regulation, explains apparent drought buffering during Chile's megadrought

Francisco Zambrano Bigiarini

2026-08-04

Much of what the hydrological literature attributes to reservoir drought buffering may be a measurable siting confound. We treat reservoir presence as a basin-level intervention in Chile's megadrought and separate siting from operation with an upstream/downstream placebo: regulated reaches below each dam are compared against unregulated reaches above, which share climate, geology, and selection history but not the reservoir. Across the streamflow pathway the apparent buffering is as strong upstream as downstream (differential slopes -0.20 versus -0.16 , randomization-inference $p = 0.39$), the signature of aridity, not regulation. A hard-balanced match

16 forcing aridity to be identical across groups (standardized mean difference
17 0.000) makes even the apparent buffering vanish entirely (intent-to-treat
18 -0.02 , $p > 0.85$). The streamflow null is well powered, excluding operational
19 buffering above 46 to 59% of baseline, and a positive control confirms that
20 the design recovers injected effects one to one. Dammed basins hold 4.5
21 times more cropland than matched controls, a siting legacy stable across the
22 2000 to 2024 record with no detected megadrought divergence. The storage
23 band drifts downward in step with the independent decline in unregulated
24 streamflow. Demand-side outcomes, cropland expansion, water rights ac-
25 cural, and evapotranspiration, are consistent with no operational effect but
26 underpowered as stand-alone tests. Through the megadrought we detect no
27 operational buffering and no megadrought-era demand expansion beyond
28 the siting legacy. The upstream placebo is a portable method for regulated
29 rivers with gauges above and below dams. Adaptation must shift toward
30 managing demand.

31 **Introduction**

32 A persistent obstacle separates what we think reservoirs do during drought from what
33 they actually do: dams are built where water is already scarce, so dammed basins
34 differ systematically from undammed ones before any drought strikes. The short-term
35 hydrological benefit of storage is documented (Wu et al. 2017, 2018), but whether
36 reservoirs buffer multi-year drought, or whether the apparent buffering is a statistical

37 shadow of where dams sit, has remained untested for want of a credible counterfactual
38 (Di Baldassarre et al. 2018). The distinction matters. If buffering is real, expanding
39 storage is a rational adaptation to intensifying drought. If it is siting, then the reflexive
40 global response, more dams, invests in the wrong mechanism.

41 The reservoir-effect hypothesis (Di Baldassarre et al. 2018) formalizes the concern: greater
42 supply invites expansion, so demand grows to meet and then exceed each increment of
43 storage (the supply-demand cycle), and prolonged reliance on stored abundance erodes
44 the incentive to adapt, magnifying the damage when drought finally outlasts the reservoir.
45 Consistent with this, global demand has outpaced storage for decades, returns of new
46 dams are diminishing (Li et al. 2023), and most reservoirs now fail to refill (Z. Wang et
47 al. 2024). Whether these feedbacks dominate, or are outweighed by the protection dams
48 provide, remains debated (Kellner 2021). But the empirical literature has a structural
49 weakness: dammed-versus-undammed or before-after comparisons cannot separate a
50 reservoir effect from the siting decision, because reservoirs concentrate where irrigation
51 demand and aridity already concentrate.

52 We address this with a within-basin falsification test. If a dam buffers drought, streamflow
53 transmission from meteorological to hydrological drought should flatten only below the
54 dam, where regulation operates. If it flattens equally at the unregulated upstream reaches
55 of the same basins, the attenuation is siting, not regulation. This upstream/downstream
56 placebo is the paper’s central identification strategy. We complement it with a matched

57 cross-basin comparison that conditions each dammed basin against hydroclimatically
58 comparable undammed basins, and we evaluate both against the drought each basin
59 actually experienced rather than against calendar time, breaking the collinearity between
60 the megadrought onset and the treatment clock.

61 Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an
62 uninterrupted run of dry years shading into structural aridification with a documented
63 anthropogenic component (R. D. Garreaud et al. 2017; R. Garreaud et al. 2025;
64 Zambrano et al. 2025; Boisier et al. 2018; Vicente-Serrano et al. 2019). The 1981 Water
65 Code makes water-use rights perpetual private property that the state cannot readily
66 curtail, so the policy reflex has been supply-side: new dams and inter-basin transfers,
67 alongside market incentives that concentrated water in high-value export crops (Barriá
68 et al. 2021; Reade Malagueño and D’Odorico 2024; Rivera et al. 2016). In the snow-fed
69 basins of central Chile demand already outruns supply in roughly three years of every
70 ten (Webb et al. 2021). Chile thus exhibits every ingredient of the reservoir effect:
71 extreme persistent forcing, a supply-side response, and demand expansion under weak
72 adaptive feedback, making it a most-likely case for detecting a reservoir effect if one
73 operates. At the same time, its short, steep Andean catchments, with streamflow gauges
74 above and below major dams, provide the physical setup the upstream placebo requires,
75 a configuration present across the world’s regulated mountain rivers from the Sierra
76 Nevada to the European Alps to the African Rift.

77 We apply this design across four outcome pathways: streamflow drought transmis-
78 sion, cropland area change, consumptive water-rights accrual, and evapotranspiration.
79 Throughout, outcomes are evaluated at annual to multi-year accumulations, the timescale
80 of structural drought, not of seasonal operation. We find that across the streamflow
81 pathway the apparent buffering is a siting confound, that the upstream placebo identifies
82 it, and that the operational effect of reservoirs on drought brackets zero once siting is
83 matched out. The upstream placebo is a portable method: wherever rivers have gauges
84 above and below dams, the same falsification test can separate siting from effect.

85 **Results**

86 **Reservoirs sit where water is already scarce**

87 Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster
88 in the arid centre and north, so a raw comparison would confuse the reservoir with its
89 location. We separate them with the design in Figure 1. First, we compare each dammed
90 basin only with undammed basins sharing its climate, size and elevation, keeping 21 of
91 24 dammed basins against an effective 91 of 244 controls (Supplementary Figs. S4 and
92 S5), and we condition on the drought each basin experienced, comparing impact per
93 unit of deficit. Second, within each dammed basin we compare regulated reaches below

the dam against unregulated reaches above it. A genuine reservoir effect must appear in these comparisons; in none does it.

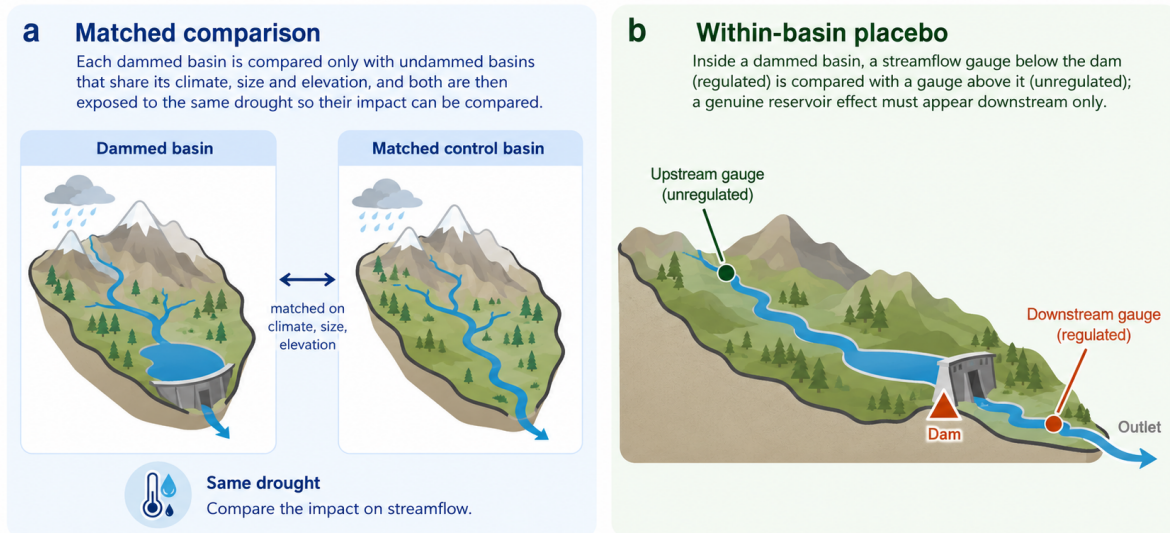


Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin is compared only with undammed basins that share its climate, size and elevation, and both are then exposed to the same drought so their impact can be compared. (b) Within-basin placebo: inside a dammed basin, a streamflow gauge below the dam (regulated) is compared with a gauge above it (unregulated); a genuine reservoir effect must appear downstream only. Schematic; the full study-area map and matching diagnostics are in Supplementary Figs. S4 and S5.

The apparent buffering is the location, not the dam

The most direct test is on streamflow. Meteorological drought (Standardized Precipitation Evapotranspiration Index of 12 months, SPEI-12) propagates into streamflow drought (Standardized Streamflow Index of 12 months, SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where

101 dams sit, not what they do. If the dam were doing the buffering, transmission would
 102 flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches
 103 of the same basins (differential slope -0.20 upstream versus -0.16 downstream; Figure 2,
 104 panel b). The national slope gap of -0.18 accordingly collapses to about -0.05 once
 105 dammed and control basins are compared within the same climate region, and in the
 106 high-supply south, where buffering should be easiest, the upstream and downstream gaps
 107 are identical (-0.036 versus -0.034). None survives randomization inference, which asks
 108 how often reshuffling the dammed label among comparable basins produces a gap this
 109 large by chance ($p_{\text{perm}} = 0.39$). Nor is the pooled slope hiding early buffering: at drought
 110 onset (2010–2014, storage near-full) the upstream and downstream gaps are identical
 111 (-0.43 both), and the later weakening is sharpest in the unregulated placebo itself,
 112 forcing, not depletion (Supplementary Table S24). The apparent buffering is therefore
 113 the between-region aridity gradient, the signature of a siting confound, not regulation;
 114 a decomposition confirms the signal dies only when each climate region has its own
 115 baseline transmission (Supplementary Fig. S3 and Table S4). Hard-balancing aridity
 116 gives the same verdict without parametric adjustment: with log-aridity fully balanced
 117 (effective control sample 19), the apparent buffering vanishes (intent-to-treat -0.02 ,
 118 upstream -0.05 , permutation $p > 0.85$; Supplementary Table S17), exactly what a siting
 119 confound predicts. The parametric adjustment it replaces interpolates within support,
 120 every dammed basin inside the control aridity range (Supplementary Fig. S8). Nor is
 121 the null a standardization artifact: SSI is normalized per gauge, which would divide

122 out dam-induced variance compression, but on log flow the gaps stay null (downstream
 123 +0.04, upstream +0.10, permutation $p \geq 0.42$) and downstream relative variability
 124 is uncompressed ($p = 0.43$; Supplementary Table S19). A direct geomorphic metric
 125 agrees: local relief, separating bedrock from alluvial reaches, does not predict control
 126 transmission, leaves the placebo unchanged, and shows no buffering among alluvial units
 127 (Supplementary Table S22). Upstream gauges sit higher, but the 0.54 km up/down
 128 elevation gap times the control transmission sensitivity (-0.21 per km, $p < 0.001$
 129 across $n = 84$ control gauges) already over-explains the observed -0.04 difference and
 130 cannot manufacture a downstream-specific effect, leaving a residual downstream-specific
 131 buffering of at most about 0.07 (13% of the 0.53 downstream baseline), well below the
 132 46 to 59% detectable. A piecewise fit around the Andean snowline (~ 2000 m) finds no
 133 break in this elevation sensitivity ($p = 0.25$; Supplementary Table S20), supporting the
 134 linear extrapolation. The placebo's identification therefore does not hinge on an untested
 135 elevation assumption: the confound is quantified, its magnitude is measured, and it
 136 cannot produce the downstream-specific buffering a genuine reservoir effect requires
 137 (Supplementary Table S2).

138 **No reservoir effect survives across any operational pathway**

139 The same answer holds across the other pathways (Figure 3, Table 1). Each effect is
 140 the differential response of dammed and matched-control basins to the same deficit,

Apparent streamflow buffering is siting, not the reservoir

Effect is permutation-null and as strong in UPSTREAM (unregulated) gauges

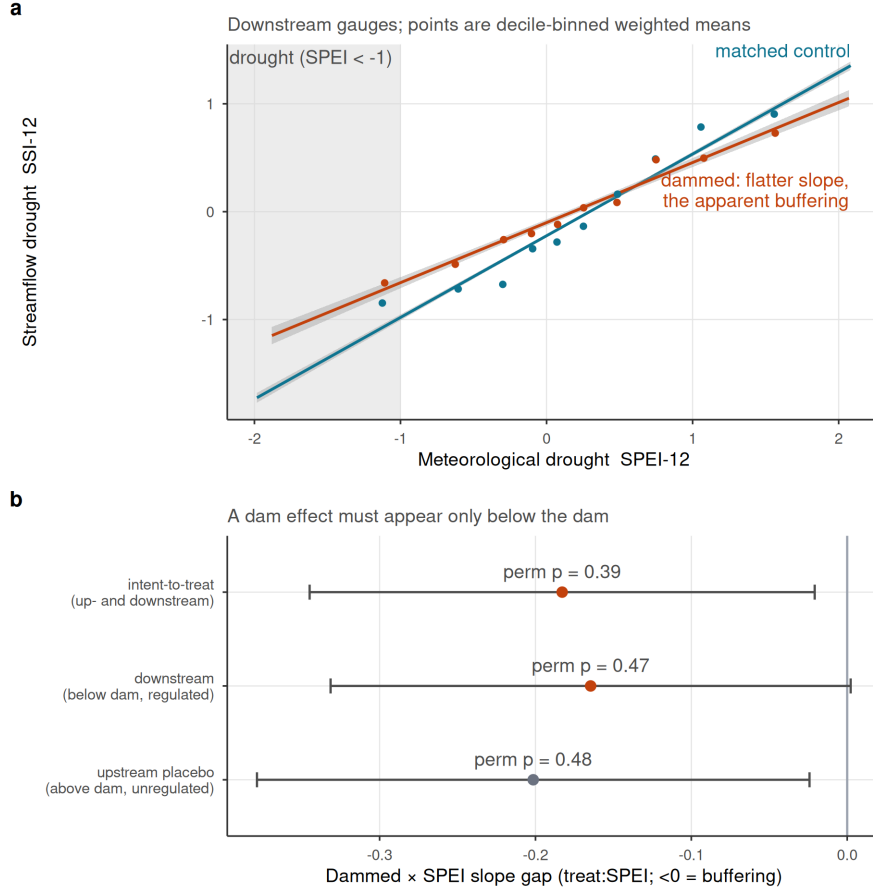


Figure 2: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate $\pm$ 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

each estimate scaled by its own standard error, so a value inside ± 2 is indistinguishable from no effect. Every estimate that carries evidential weight brackets zero: cross-sectional cropland-area expansion (-0.69 ha km^{-2} , 95% CI $[-2.4, 1.1]$) and its forcing-conditioned slope gap ($p_{\text{perm}} = 0.91$). The evapotranspiration pathway is relegated to the Supplementary Information: whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the post-hoc orchard stratum relies on MODIS (Moderate Resolution Imaging Spectroradiometer) MOD16 product, whose documented bias toward zero over irrigated land, a product property identified independently of our results, disqualifies its null (Supplementary Table S21, Fig. S7).

The null is informative but its power is concentrated in the streamflow pathway. With about 21 dammed clusters the design cannot claim exact equivalence to zero but rules out buffering above 46 to 59% of baseline, and a positive control confirms it: injected buffering is recovered one-to-one and detected beyond that threshold (Methods; Supplementary Fig. S2 and Tables S1, S3). Of the remaining pathways, only the streamflow null excludes material effects at operationally relevant magnitudes. The cropland and evapotranspiration designs are far weaker, with minimum detectable effects orders of magnitude above baseline (7,330% and 237% of baseline, respectively; Supplementary Table S2); their nulls corroborate rather than independently establish the absence of an effect. The water-rights count is comfortably null under all inference frameworks ($p_{\text{perm}} \geq 0.42$), while the winsorized volume is a more nuanced result discussed below.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted DiD slope gaps (treat:SPEI). The DiD row and water-rights accrual are judged by randomization-inference p-values (p), the design’s primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. Evapotranspiration outcomes are relegated to the Supplementary Information (Table S21, Fig. S7): whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the orchard rows rely on MOD16, whose documented underestimation over irrigated land biases them toward zero.

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Cropland-area expansion	ha km ⁻² added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Water-rights accrual	rights / 100 km ⁻² added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD (slope-gap)	Cropland area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null

Reservoirs mark prior vulnerability; no megadrought divergence is detected

Dammed basins carry the vulnerability reservoirs are blamed for concentrating: about 4.5 times more cropland than matched controls (10.8% versus 2.4% of basin area; Figure 4, panel a). This gap is a *siting* signature, stable across the 2000–2024 record. A dynamic event study finds the differential trend flat before 2010 (cluster-robust $p = 0.65$, permutation $p = 0.72$) with no post-2010 divergence detected ($p = 0.81$; Figure 4, panel b). Both are bounded nulls, not confirmations of parallelism: the pre-trend test detects only divergence accumulating to about half the control cropland level by 2009 (the

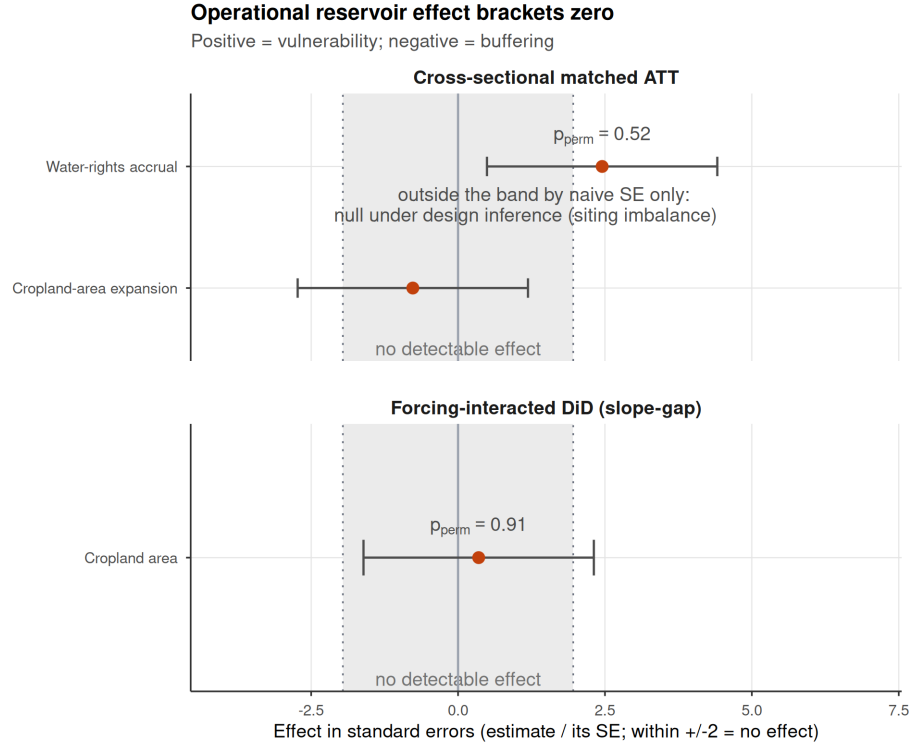


Figure 3: Dammed-versus-control effect across the estimators, each expressed uniformly in its own standard errors (estimate divided by its SE) so the axis metric is identical for every row; an estimate inside ± 2 is indistinguishable from no effect. Rows are grouped by inference regime: cross-sectional matched ATTs (top) and forcing-interacted DiD slope gaps (bottom). Bars span ± 1.96 SE (the 95% reference interval) and dotted lines mark ± 1.96 ; a positive effect would indicate induced-demand vulnerability, a negative one operational buffering. Rows judged by randomization inference are annotated with their p (the design's valid inference at ~ 21 clusters, since the ± 1.96 SE guides over-reject there). Under that valid inference every estimate brackets zero. Evapotranspiration outcomes are relegated to the Supplementary Information (Table S21) as product-limited and confounded. Water-rights accrual sits outside ± 1.96 by its naive standard error yet is null by the design ($p_{\text{perm}} = 0.52$; the apparent induced demand lies inside the permutation null expected from the dammed basins' unbalanced aridity, Supplementary Fig. S6).

170 observed trend is a seventh of that), and the permutation envelope widens after 2015, so
171 the late panel cannot rule out substantial divergence (Supplementary Table S14). The
172 null is metric-invariant: on log cropland share the slope gap is likewise permutation-null
173 ($p = 0.37$; Supplementary Table S20). Reservoirs mark where cropland already was;
174 within these bounds they neither expanded it nor changed its drought response.

175 The same holds for the most direct demand measure. In the Chilean Water Directorate
176 (DGA) registry of consumptive water rights, dammed basins appear to accrue rights
177 faster than controls, on both the outlier-robust count (+20 per 100 km²; Figure 3,
178 Table 1) and the winsorized volume (+2.5 l/s per km²; Supplementary Fig. S6), each
179 with an analytic interval excluding zero. This is the naive-analysis trap the design
180 catches: dammed basins are more arid, arid basins carry denser water-rights activity,
181 and the imbalance drives the gap. The count does not survive randomization inference
182 ($p_{\text{perm}} = 0.52$; Supplementary Fig. S6) and remains null under a fully non-parametric
183 match on the aridity-overlap subset ($p_{\text{perm}} = 0.42$; Supplementary Table S7). The
184 winsorized volume is more delicate: stable across thresholds and physically validated
185 (Supplementary Table S8), its overlap-subset permutation p falls to 0.005, but the
186 residuals are spatially autocorrelated (Supplementary Table S9) and under the governing
187 cuenca-block permutation it is not significant ($p_{\text{perm}} = 0.077$, cuenca-clustered interval
188 $[-0.3, 4.1]$). Ranked raw, uncapped volumes give the same verdict: the design rank
189 contrast is null under cuenca-block permutation ($p = 0.15$; Supplementary Table S18).

190 The water-rights evidence is therefore not uniformly null: the count is cleanly so under all
 191 frameworks, the ranked raw volumes are null under spatial inference, and the winsorized
 192 volume sits closer to the boundary, with a unit-level signal ($p = 0.005$) that the spatial
 193 structure partially explains but does not fully erase (cuenca-block $p = 0.077$). The gap
 194 between naive and spatial inference flags the spatial concentration of large consumptive
 195 rights in dammed cuencas, a pattern the design can bound but whose attribution to
 196 siting versus reservoir-induced demand it cannot fully resolve at this spatial grain. No
 197 induced demand is robustly attributable to the reservoir once siting and spatial structure
 198 are accounted for, but the volume signal warrants further investigation with parcel-level
 199 consumption data.

200 **Storage falls; a change in refill amplitude cannot be resolved**

201 If the streamflow pathway excludes material operational buffering and the demand-side
 202 pathways are consistent with no amplification at their respective detection limits, what
 203 matters is how much water there is to store (Figure 5). Over 2005–2024 the storage
 204 band drifts downward in step: peak storage falls by 1.3 percentage points of capacity per
 205 year ($p = 0.013$) and the trough almost identically (1.2 pp yr^{-1} , $p < 10^{-4}$), while the
 206 seasonal amplitude shows no detectable trend (-0.06 pp yr^{-1} , $p = 0.91$; Supplementary
 207 Table S5). That amplitude interval is wide: whether refill amplitude, the buffering
 208 capacity, has degraded cannot be determined, and we draw no conclusion about it.

Reservoirs do not alter irrigated-area dynamics

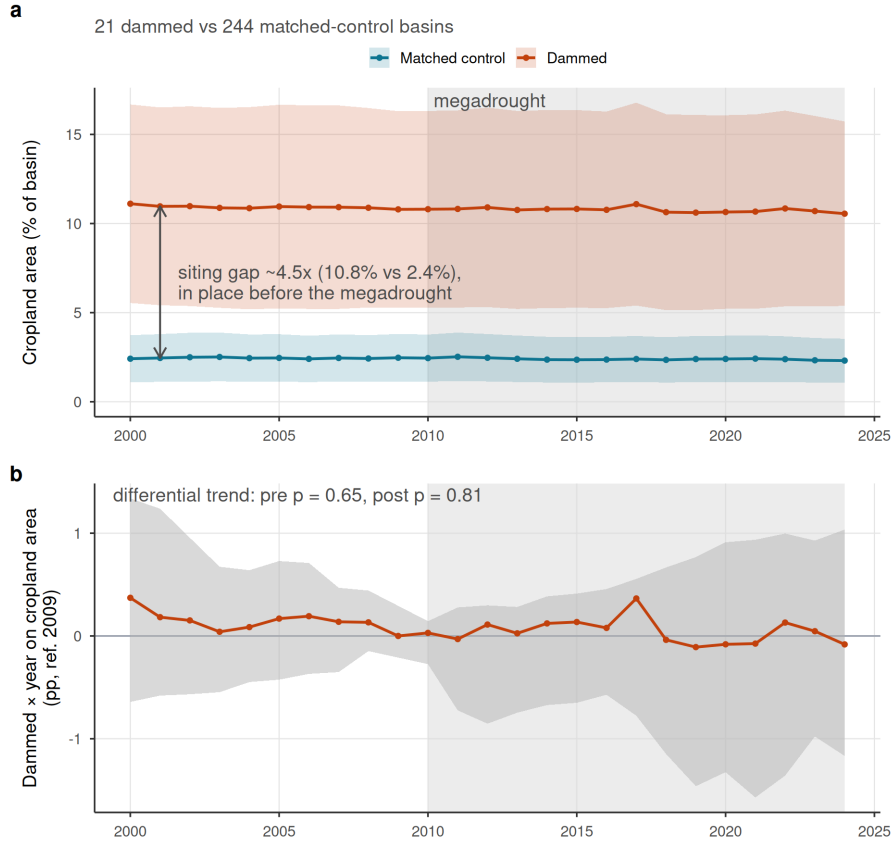


Figure 4: Observed cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands (weighted mean $\pm$ 1.96 SE across basins); the y-axis is anchored at zero. Dammed basins hold $\sim 4.5\times$ more cropland (10.8% vs 2.4% of basin area) but the trajectories evolve in parallel. (b) Dynamic event study (dammed $\times$ year, reference 2009): the grey band is the 95% randomization-inference null envelope (2.5th to 97.5th percentile of the dammed $\times$ year coefficients under treatment permutation within Köppen $\times$ aridity strata); coefficients inside the band are indistinguishable from no divergence, and cluster-robust intervals are omitted as anti-conservative at ~ 21 treated clusters. The differential pre-trend is flat (pre-2010 trend-slope cluster-robust $p = 0.65$, permutation $p = 0.72$) and no post-2010 divergence is detected (post $p = 0.81$); a linear trend-slope test is used because the joint event-study Wald over-rejects at ~ 21 treated clusters. The null envelope widens after 2015, reflecting declining power in the later panel, so the flat post-2015 trajectory is a bounded null: divergence smaller than the envelope cannot be ruled out. The 2010-2024 megadrought is shaded in both panels.

209 The decline is a level shift at the megadrought onset rather than a secular line (peak
210 87% of capacity before 2010, 67% after; trough 40% to 25%; Supplementary Table S5).
211 That decline is commensurate with the supply reduction measured independently in
212 the matched controls: unregulated control streamflow fell about 2.9% per year over
213 2005–2020 (dammed 3.4%), consistent with documented megadrought runoff deficits (R.
214 D. Garreaud et al. 2017; Alvarez-Garreton et al. 2021) and comparable to the ~1.4% per
215 year decline in peak storage. Storage is a mass balance, so the band alone cannot separate
216 falling inflows from sustained releases; the supply reading rests on that independent
217 streamflow decline. Our direct demand outcomes point the same way: neither cropland
218 area nor water-rights allocation shows expansion (Figure 4, Table 1; Supplementary Fig.
219 S6) (Li et al. 2023; Z. Wang et al. 2024).

220 Discussion

221 Across a matched national design, multiple estimators, and a within-basin falsification
222 test, the apparent drought buffering attributed to reservoirs is a measurable siting
223 confound. The upstream placebo is the decisive evidence: a regulation effect must appear
224 only below the dam; it does not, and what remains is the aridity gradient. Dammed
225 basins hold about 4.5 times more cropland than matched controls, but that gap is fixed
226 across the record, with parallel pre-trends and no detected megadrought divergence. One

Storage declines; a change in refill amplitude cannot be resolved

The whole band drifts down; the peak-trough amplitude trend is inconclusive (wide interval)

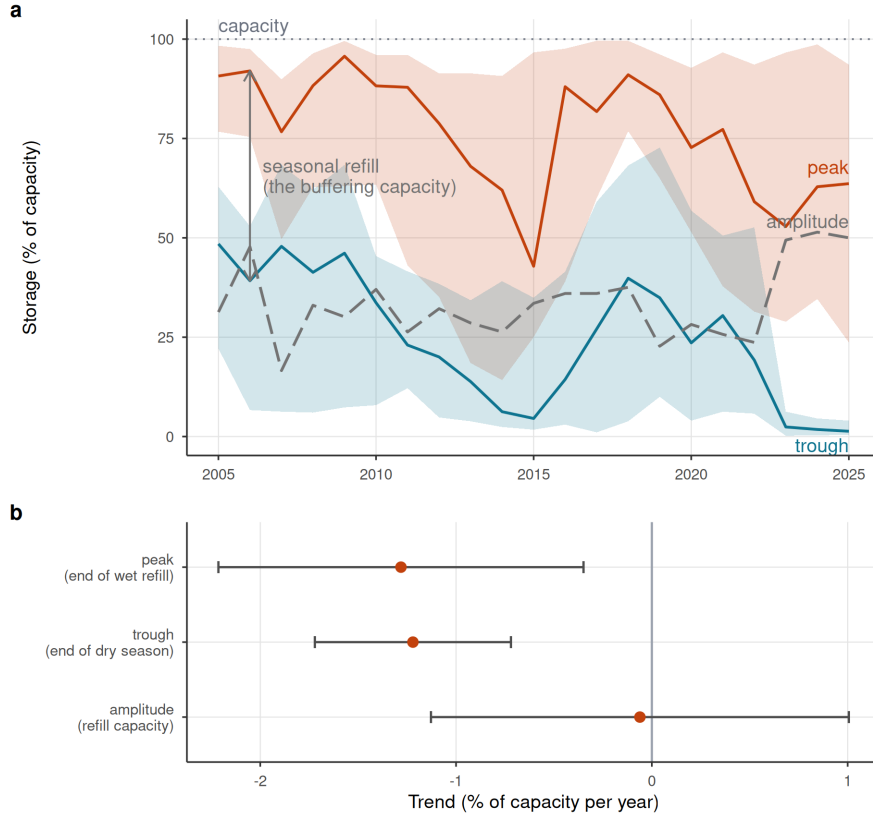


Figure 5: Reservoir storage declines; a change in seasonal refill amplitude cannot be resolved. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid lines) with interquartile ribbons across reservoirs, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the whole band drifts down over 2005-2024 while the amplitude does not narrow. Central lines are the realized medians, not linear fits; the net trend is quantified in (b). (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate ± 1.96 cluster-robust SE): peak and trough both decline while the amplitude trend is not distinguishable from zero (a wide interval, so statistically unchanged rather than proven constant), consistent with a supply-side level decline rather than refill/buffer degradation. Descriptive: pooled within-reservoir trend, no control group.

227 reading fits: reservoirs mark where water-dependent vulnerability already concentrated
228 rather than make it. This is a finding, not a failure of power: the apparent effect has an
229 identified cause and the positive control shows the design detects buffering when it is
230 present.

231 The corollary is transferable: dammed-versus-undammed or before-after contrasts inherit
232 the full siting confound (López-Moreno et al. 2009; Wu et al. 2017; Schilstra et al. 2024;
233 Lee, Kang, and Kim 2026). The upstream placebo ports to regulated rivers with gauges
234 above and below a dam, most directly to short, steep catchments like Chile’s; in large
235 low-gradient basins, floodplain storage and transmission losses would blur the contrast;
236 porting it would need routing-corrected lags or naturalized-flow counterfactuals.

237 If the streamflow pathway excludes material operational buffering at multi-year scale
238 and the demand-side outcomes are consistent with no amplification at their respective
239 detection limits, what matters is the water available to store. The storage band steps
240 down at the megadrought onset, peak and trough together, while the amplitude trend is
241 too imprecise to determine whether buffering capacity degraded. The powered result
242 is the level decline, the expected mass-balance outcome of persistent inflow deficits;
243 descriptive and confounded with management change, it supports no causal attribution
244 but matches the independent decline in unregulated control streamflow. Read against
245 Chile’s structural aridification (Zambrano et al. 2025; R. D. Garreaud et al. 2017;
246 Alvarez-Garreton et al. 2021), accelerating Andean warming and snow loss now cutting

247 into the snow-fed streamflow these basins depend on (Kim, Rajaram, and Lakshmi 2025),
248 and the global pattern of reservoirs failing to refill (Z. Wang et al. 2024; Li et al. 2023),
249 the binding constraint is shifting toward supply.

250 These findings give the contested socio-hydrological *reservoir effect* and *supply-demand*
251 *cycle* (Di Baldassarre et al. 2018; Kellner 2021) a credibly identified test, and find
252 the protective function over-attributed to storage. Because treatment is time-invariant,
253 we test the spatial imprint of these feedbacks, not the longitudinal loop. The 4.5-
254 fold cropland gap could be the equilibrated legacy of demand induced at construction,
255 so our null covers the megadrought era; historical induced demand is testable only
256 with staggered commissioning. And the record opens in the scarcity phase, after any
257 expansion under perceived abundance, so it bounds drought-era expansion, not the cycle
258 as a whole. Within scope the nulls refine the theory: the cycle leaves no operational
259 imprint during drought, so its mechanisms likely act through siting and construction
260 rather than operation, a distinction current models omit. How far the demand-side
261 null travels depends on governance: the siting lesson holds wherever dams are placed
262 non-randomly; prior appropriation, which forfeits unused rights, may amplify induced
263 demand while adaptive allocation curtails it. Whether the null is general or the mark of a
264 fully allocated market we cannot adjudicate: rights kept accruing in dammed basins, but
265 systems with water still unallocated could express an induced demand that ours cannot.
266 The upstream placebo itself ports more broadly. Short, steep catchments with gauges

above and below dams are common across the world’s regulated mountain rivers: the California Coast Ranges and Sierra Nevada, the Spanish Sierra Nevada and Pyrenees, the Italian Apennines and Alps, the African Rift, and the Andean cordillera beyond Chile. In large low-gradient basins, the Murray-Darling, the Colorado below Lake Mead, or the Indus plains, floodplain storage and transmission losses would blur the above/below contrast and routing-corrected lags or naturalized-flow counterfactuals would be needed. The placebo method is therefore not universal, but its scope conditions are explicit and its required data, paired gauges above and below dams, exist across much of the world’s irrigated mountain-front hydrology.

Chile’s 1981 Water Code makes water-use rights perpetual private property, so the reflexive response has been supply-side, new dams and inter-basin transfers (Barriá et al. 2021; R. Garreaud et al. 2025); the 2022 reform moves toward the adaptive allocation our results argue for (Macpherson et al. 2023) but postdates our window. Under structural aridification, building more storage is a supply-blind response to a supply problem: the buffering dams provide is a property of where they sit, not of what they do (Ho et al. 2017). Adaptation must shift toward demand management, reallocation and managed aquifer recharge (Grafton et al. 2018; Scanlon et al. 2023; Rivera-Vidal et al. 2025).

Several limitations bound these conclusions, each quantified in the Supplementary Information. The result is a bounded null: the design excludes streamflow buffering above 46 to 59% of baseline, demand-side outcomes are far less powered and the pre-trend

test power-limited (Supplementary Tables S2, S14), so causal weight rests on the placebo,
demand nulls corroborating; all survive the hard-balanced match (Supplementary Table
S17). The registry measures paper rights, not wet water: Chilean basins are widely
over-allocated and granted rights can sit unused or hoarded (Barría et al. 2021; Reade
Malagueño and D’Odorico 2024), so under drought, when physical availability binds,
the accrual we test is an entitlement response that non-use could dampen; cropland
is the physical check. Both metrics also miss intensification within an unchanged
footprint; central Chile’s documented shift toward thirsty export fruit (Reade Malagueño
and D’Odorico 2024) is exactly such a transition, invisible to both; a consumption-
grade test awaits thermal ET or crop-transition maps. Groundwater substitution is
bounded, dammed and control water tables falling at indistinguishable rates across 213
DGA wells despite severe regional drawdown (Taucare et al. 2024); monitored wells
need not sample the orchard footprints where pumping concentrates, so unmonitored
drawdown could sustain a historically reservoir-induced footprint (Supplementary Table
S10). In that reading the reservoir effect is not absent but displaced to the aquifer,
a construction-era legacy persisting on a pool the design cannot see; the well bound
and flat cropland trajectories argue against substitution at scale but do not close the
pathway. The snowmelt confound, acute in these snow-fed Andean basins (Ayala et
al. 2025; Götte and Brunner 2024), is tested on the differential estimand and excluded
across four specifications (Supplementary Table S12). The geomorphic check leaves
the placebo unchanged, though 2.5 km relief cannot resolve aquifer storage itself, a

308 residual caveat (Supplementary Table S22). The placebo also spans land-cover and
 309 abstraction contrasts, downstream valleys more cultivated, upstream reaches carrying
 310 unrecorded diversions; gauge fixed effects absorb these level differences, and drought-
 311 intensified upstream abstraction would steepen upstream transmission, working against
 312 the observed upstream attenuation, not producing it. Administrative closure cannot
 313 force the demand null: treated basins entered the drought with higher allocated stock
 314 yet accrued rights 2.7 times faster, and parallel control cropland contradicts collapse
 315 prevention (Supplementary Table S15). No matched pair shares a watershed; outbound
 316 transfers would shrink the gap and imports would mimic downstream-specific buffering,
 317 which is absent (Supplementary Table S15); irrigation-only restriction leaves the result
 318 unchanged (Supplementary Table S11). MOD16's bias toward the null excludes the
 319 ET pathway from evidence (Supplementary Table S21); the demand conclusion rests
 320 on cropland and the registry, with thermal ET as future work. Finally, the 12-month
 321 accumulation targets multi-year vulnerability, not the sub-seasonal buffering reservoirs
 322 also provide, which we neither test nor dispute. The fleet is partly seasonal by design,
 323 the median treated reservoir storing 0.3 years of downstream flow, yet seven of 17 hold
 324 at least half a year, four exceed a year, and none empties seasonally (Supplementary
 325 Table S23): carryover was physically available, so its absence in outcomes is not a design
 326 tautology. Two extensions can test what our window cannot: a staggered event study
 327 of cropland around historical commissioning would probe construction-phase induced
 328 demand; a rule-curve quasi-experiment on release records, once public, would isolate

329 operation.

330 At the multi-year timescale of structural aridification, then, the design detects neither
331 operational buffering nor megadrought-era demand expansion beyond the siting legacy:
332 the apparent buffering is a statistical shadow of where dams sit, not of what they do,
333 while the water available to store declines. The nulls are bounds, not zeros; effects inside
334 them can be volumetrically material (Supplementary Table S16). But under structural
335 aridification with no operational buffering at the multi-year scale, building more storage
336 is a supply-blind response to a supply problem. Chile is currently evaluating at least
337 26 new reservoir sites to expand storage, while its agricultural water demand continues
338 to grow under perpetual private water rights that the 2022 Water Code reform only
339 begins to address (Macpherson et al. 2023). The alternative is not theoretical: demand
340 reallocation through active water markets, irrigation efficiency, managed aquifer recharge,
341 and crop transitions toward lower water footprints are available now and do not depend
342 on rainfall (Grafton et al. 2018; Scanlon et al. 2023; Rivera-Vidal et al. 2025). Resilience
343 to a falling supply ceiling will come from confronting demand and allocation, not from
344 more impoundments.

Methods

Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of dammed versus undammed sub-watersheds, with reservoir presence treated as a basin-level intervention. Because the reservoirs in most treated basins (18 of the 24) were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting* (Di Baldassarre et al. 2018).

Spatial units and treatment assignment

The analysis grain is the DGA sub-watershed (467 units), chosen over the coarser watershed grain (101 units) for power and common support, reservoirs sit in Chile's largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (detailed in the archived analysis repository). A sub-watershed is treated if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving 24 treated units (two reservoirs share a sub-watershed). After entropy balancing, three high-Andes treated units fall outside local elevation common support, leaving 21 treated basins in all analyses; the progression from 26 reservoirs to 24 treated sub-watersheds to 21 retained treated units is fixed across every estimator and table. The reservoirs are predominantly irrigation (20 of 26; three hydropower, two potable supply, one mixed), so the treated set is dominated by the use type built to buffer agricultural drought, and the headline streamflow result is unchanged when restricted to irrigation-only treated basins (Supplementary Table S11). An untreated sub-watershed that lies within a watershed containing a reservoir is flagged as contaminated (it shares the regulated main stem and is not a clean counterfactual) and removed (45 units), leaving 398 clean controls. This contamination rule defuses the most direct downstream-spillover (SUTVA: Stable Unit Treatment Value Assumption) violation; residual adjacency and shared-aquifer spillovers between neighbouring sub-watersheds are carried as an identification caveat and addressed with watershed-level checks: where spatial dependence is present, inference is re-run

with watershed-block permutation and watershed-clustered intervals (Supplementary Table S9), and no matched treated-control pair shares a watershed (Supplementary Table S15).

Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (the DGA record runs January 2005 to April 2026; we analyse the complete calendar years 2005–2024) with a per-reservoir capacity column (maximum level in hm^3); storage and capacity share units (hm^3), so percent-of-capacity ($100 \times \text{level} / \text{maximum level}$) is the cross-reservoir-comparable state variable. Static attributes (commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. The Standardized Precipitation-Evapotranspiration Index (SPEI) at a 12-month accumulation, derived from Climate Hazards Center InfraRed Precipitation version 3 (CHIRPS) and Climate Hazards Center InfraRed Temperature with Stations (CHIRTS) and standardized on the 1991–2020 climatological baseline (Sergio M. Vicente-Serrano, Beguería, and López-Moreno 2010), is the primary exogenous dose; CHIRPS-derived standardized indices are validated for Chilean drought monitoring, including ENSO-modulated variability across the country (Zambrano et al. 2017; Meza 2013; Oertel, Meza, and Gironás 2020). We use the 12-month accumulation because it

400 matches the annual, snowmelt-fed hydrological cycle of central Chile (winter precipitation
401 stored as snow and released through the growing season) and so captures the inter-
402 seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which
403 multi-year vulnerability (rather than sub-seasonal operation) is expressed. The treatment
404 estimates are insensitive to this choice: re-estimating the forcing-conditioned effect
405 with 6- and 24-month accumulations leaves it materially unchanged (archived analysis
406 code). By construction the 12-month window does not resolve short-term (1–3 month)
407 operational buffering, which reservoirs are documented to provide (Wu et al. 2017, 2018)
408 and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing
409 (estimating effects on the deficit-to-impact *slope* rather than on levels) is how we break
410 the collinearity between the 2010-onset megadrought and the treatment clock.

411 **Hydrological drought.** Daily streamflow from the DGA network (2000–2020), gathered
412 by the CR2 (Center for Climate and Resilience Research, <https://cr2.cl>) is aggregated to
413 the Standardized Streamflow Index at 12 months [SSI-12; Sergio M. Vicente-Serrano et
414 al. (2012); Lema et al. (2025)], the regulated-flow analogue of SPEI-12 and the index in
415 which meteorological deficits are re-expressed as hydrological drought, a propagation that
416 reservoir operation is known to modify by lengthening and attenuating the transmitted
417 signal (Van Loon 2015; W. Wang et al. 2016), used as the outcome in the within-basin
418 placebo (below). Gauges are classified as upstream or downstream of each dam from
419 an SRTM digital elevation model. Because SSI standardizes each gauge on its own, for

regulated gauges post-dam, flow distribution, the transmission slope tests distribution-
relative propagation and would divide out any dam-induced variance compression; we
therefore re-estimate every buffering model and the placebo on log 12-month mean
flow, a volumetric semi-elasticity (percent of flow per SPEI unit) that no per-gauge
normalization can compress, and test the compression premise itself, the deseasonalized
relative flow variability by gauge class (Supplementary Table S19).

Ecological and land-use outcomes. (i) annual cropland area from MapBiomass
Chile (<https://chile.mapbiomas.org/>) Collection 2 (30 m, 1999–2024); (ii) actual evap-
otranspiration from MODIS MOD16 (8-day, ~500 m, 2001–2024), for whole basins
and for an orchard stratum delineated from the *Centro de Información de Recursos*
Naturales (CIREN) and *Oficina de Estudios y Políticas Agrarias* (ODEPA) cadastre
(<https://icet.odepa.gob.cl/>). Satellite vegetation and evapotranspiration signals track
drought-driven losses of agricultural productivity in Chile closely enough to serve as
impact outcomes (Zambrano et al. 2018).

Water rights. The DGA consolidated registry of water-use rights (*Derechos de*
Aprovechamiento de Aguas), giving for each right its grant date, consumptive/non-
consumptive type, surface/groundwater nature, use, allocated flow (harmonized to l/s),
and UTM capture point. We assign consumptive rights to sub-watershed by point-in-
polygon and build a per-basin annual panel of the cumulative allocated-rights stock,
our direct demand-side outcome. Because the registry contains severe volume errors (a

single physically impossible entry can dominate a basin’s total, e.g. a caudal exceeding 200,000 m³/s), the outlier-robust primary measure is rights *density* (count per 100 km²); a volume series in which each individual consumptive flow is winsorized at the 99.5th percentile of the consumptive-flow distribution gives the same result, and the volume null is insensitive to the winsorization threshold across the 95th, 99th, 99.5th and 99.9th percentiles (permutation p from 0.39 to 0.21; Supplementary Table S8). Both are reported (Supplementary Fig. S6). Because volumes are extremely concentrated (the largest 1% of raw entries, dominated by the data errors, hold 99.7% of registry volume), we additionally report rank-based tests on the raw, uncapped volumes: the per-right volume distribution in dammed versus control basins, and a design rank contrast (ebal-weighted mean-rank difference) under both unit-level and watershed-block permutation, so no cap or outlier can drive the volume conclusion (Supplementary Table S18).

Snow water equivalent. ERA5-Land monthly snow-water-equivalent (SWE, 0.1°, 2000–2026), reduced to per-gauge and per-sub-watershed peak-SWE climatologies plus annual peak-SWE anomalies, to test the snowmelt confound on the placebo estimand itself (Supplementary Table S12): we refit the up- and downstream buffering models with a forcing-by-snowpack adjustment, within the low-snow half of units (below-median peak SWE), across early and late halves of the panel, and with a snow-year interaction that asks whether the apparent buffering strengthens in high-snow years. ERA5-Land underestimates absolute Andean snowpack, a product limitation analogous to the MOD16 ET

460 bias we disqualify elsewhere. Both biases can attenuate estimated effects. The difference
461 in treatment is substantive rather than asymmetric: MOD16’s underestimation is land-
462 cover-dependent (concentrated over irrigated land) and potentially non-multiplicative,
463 biasing the orchard-ET slope gap toward zero in a way standardization cannot remove,
464 so we exclude those outcomes from evidence. ERA5-Land’s snow bias is predominantly
465 multiplicative and land-cover-independent, so standardization, which is insensitive to
466 multiplicative bias, removes it; all snow terms therefore enter standardized, and gauge-
467 level sensitivity checks among undammed controls are reported per standard deviation
468 of peak SWE.

469 **Groundwater levels.** DGA well hydrographs of depth to water (2000–2021; 216 wells
470 carry usable hydrographs nationally, of which 213 fall inside the matched basins and enter
471 the bound; a Hampel filter removes measurement spikes), assigned to sub-watershed by
472 point-in-polygon from well coordinates. Each well’s megadrought (2010–2021) depletion
473 rate is the ordinary least square slope of depth to water on time (m yr^{-1} , positive = falling
474 water table); per-basin values aggregate well rates by their outlier-robust median. These
475 bound the groundwater-substitution confound as a matched dammed-versus-control
476 Average Treatment Effect on the Treated (ATT) on the depletion rate (Supplementary
477 Table S10).

478 **Matching covariates.** Baseline aridity (long-term mean annual precipitation divided
479 by potential evapotranspiration, P/PET , over the 1991–2020 World Meteorological

Organization normal; PET is the atmospheric evaporative demand and P/PET is used throughout), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per sub-watershed by zonal statistics. Because the 1991–2020 normal overlaps eleven megadrought years, baseline aridity could in principle absorb post-treatment variation; we verify the window is innocuous. Unit aridity recomputed on the strictly pre-drought 1991–2009 window is essentially identical to the full-window covariate (Pearson $r = 0.998$, Spearman $\rho = 0.999$ across the 265 matched units, and 99% of units keep the same aridity tercile used for the permutation strata), and the megadrought change in P/PET does not differ significantly between treated and weighted control basins (-0.04 , $p = 0.10$, an order of magnitude smaller than the ~ 0.5 treated-control aridity gap the matching addresses), so the matching does not lean on drought-period or reservoir-influenced variation (Supplementary Table S13).

Matched control construction

We estimate the average treatment effect on the treated (ATT) using *entropy balancing* exact-matched within Köppen main group, balancing $\log(\text{area})$ and *mean elevation* (see the archived analysis repository). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size (ESS) from 9 to 91 while balancing latitude for free.

499 Entropy balancing is a calibration estimator: given the weights, covariate balance
 500 is exact by construction and requires no independence assumption across units, so
 501 spatially autocorrelated covariates affect inference rather than balance, and inference is
 502 handled at the outcome stage with watershed-block permutation and clustered intervals
 503 (Supplementary Table S9); the exact matching within Köppen main group is itself
 504 a coarse spatial blocking, and latitude, though untargeted, balances as a by-product
 505 (Supplementary Fig. S5). The procedure retains 21 of 24 treated sub-watersheds (three
 506 high-Andes units fall outside local elevation common support) and achieves essentially
 507 perfect balance (standardized mean differences (SMD) for $\log(\text{area})$ and elevation fall
 508 from 1.18 / 0.37 to 0.000, variance ratios ≈ 1), with an ESS 91 out of 244 in-window
 509 controls. The 244 in-window controls are the subset of the 398 clean controls that
 510 survives the common-support trim shared by every matching estimator: a control must
 511 lie in a Köppen main group containing a treated unit and within 250 m of that group's
 512 treated elevation range, and groups with fewer than ten such controls are dropped, so
 513 154 clean controls fall outside this support and never enter the matching pool. Baseline
 514 aridity is a genuine confound (dammed basins are markedly more arid: median P/PET
 515 0.26 vs 0.80 unweighted) but is *not* used as a balance target, because hard-balancing it
 516 collapses the ESS from 91 to 19; instead the weighting reduces the aridity imbalance
 517 to a residual SMD 0.17 (still above the 0.1 target), which the doubly-robust outcome
 518 model absorbs. We nonetheless report the strict hard-balanced fit (entropy balancing
 519 targeting log-aridity as well, SMD 0.000, ESS 19) as a primary sensitivity: with no

parametric extrapolation across the regime gap at all, the streamflow, cropland, and water-rights results remain null, and even the *apparent* buffering vanishes, the direct signature of an aridity confound rather than a reservoir effect (Supplementary Table S17). At ESS 19 the hard-balanced fit is underpowered as a stand-alone estimator; its evidential value is agreement with the powered primary, not independent power. The parametric log-aridity adjustment is itself validated among undammed controls: their baseline SPEI-to-SSI slope varies smoothly and monotonically across aridity terciles (0.65 arid, 0.70 mid, 0.74 humid), with no regime break for a linear adjustment to misfit (Supplementary Table S20). Because this residual is real, the outcome-model adjustment for aridity is *load-bearing* rather than cosmetic; the gap it bridges is one of distributional density, not of support: all 21 dammed basins lie inside the control aridity range and inside the control 5th to 95th percentile band, and the control density is strictly positive across the treated support, so the adjustment interpolates within overlap rather than extrapolating beyond it (Supplementary Fig. S8). Accordingly, the reported cross-sectional ATTs use a *linear* log-aridity adjustment, adding a quadratic aridity term leaves the qualitative conclusions unchanged (archived robustness), and restricting to the aridity-overlap subset (all 21 treated basins and 158 of 244 controls fall inside it) leaves the streamflow transmission-slope gap essentially unchanged (-0.17 vs -0.18) while shrinking the water-rights ATT to a value that now straddles zero, so the parametric adjustment is not masking a true effect (Supplementary Table S6). To rule out that the nulls are artefacts of this parametric functional form, we additionally re-fit entropy

balancing *within* the aridity-overlap subset and estimate every cross-sectional outcome with the weighting-only estimator, which carries no outcome-model aridity term and so performs no extrapolation across the regime gap. On this fully non-parametric matching specification, cropland-area expansion, orchard-ET buffering and the water-rights count all remain null (Supplementary Table S7); the winsorized water-rights volume, which behaves differently here, is treated under spatial inference below. Crucially, the study's decisive evidence, the within-basin upstream/downstream placebo, the within-region transmission-slope comparison, and randomization inference, does not use the outcome-model aridity adjustment at all, so the central nulls do not hinge on extrapolating across the aridity gap. Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most extreme large/high basins, so all treatment effects are reported with and without those units.

Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline *doubly-robust (DR)* estimator (ebal weights and an outcome model in $\log(\text{area})$, elevation, and $\log(\text{aridity})$), so functional-

form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (eбал weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin *transmission slope*, the regression of an annual outcome (e.g. log evapotranspiration) on annual SPEI-12 over 2005–2024, a standardized, cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

Table 1 carries two cross-sectional matched ATTs, both level-change outcomes rather than transmission slopes: (i) *cropland-area expansion*, the change in MapBiomass cropland area per basin over the window (ha km⁻² added, as Table 1 states), whose protection against differential megadrought exposure comes from the forcing-interacted difference-in-differences below; and (ii) *water-rights accrual*, described under Water rights above. A third cross-sectional outcome, *reservoir ET buffering*, applies the transmission-slope machinery to log evapotranspiration on SPEI over *irrigated orchard cells* (MOD16 ET on the CIREN/ODEPA cadastre footprint); because MOD16 underestimates ET over irrigated land, biasing that slope gap toward zero, the ET outcomes are reported with caveats in the Supplementary Information (Table S21) and carry no evidential weight.

579 **Forcing-interacted difference-in-differences.** On the observed area and ET panels
580 we fit

$$y_{it} = \beta (\text{dammed}_i \times \text{SPEI}_{it}) + \gamma \text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

581 with sub-watershed (α_i) and year (τ_t) fixed effects and entropy-balancing weights.
582 Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar
583 year: the year fixed effects absorb the common megadrought shock, and β is the differential
584 deficit-to-impact slope between dammed and matched-control basins, robust to the siting-
585 in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability
586 hypothesis predicts $\beta > 0$.

587 **Equivalence testing and minimum detectable effect (MDE).** Because the headline
588 results are nulls, we move beyond failure-to-reject and bound the operational effect.
589 For each null we report a two one-sided test (TOST): the effect is declared equivalent
590 to zero if its 90% confidence interval lies within a negligible region of $\pm 25\%$ of the
591 baseline (untreated) transmission slope. We are not aware of an established hydrological
592 or operational standard defining a “negligible” reservoir effect on drought transmission,
593 so we anchor the margin in the granularity of operational drought classification rather
594 than choosing it arbitrarily. Standardized drought indices are managed in discrete
595 severity categories each 0.5 standardized units wide (McKee, Doesken, and Kleist

1993; Sergio M. Vicente-Serrano, Beguería, and López-Moreno 2010). Because the
 estimand is a slope, the absolute change it induces in the propagated index scales
 with the SPEI deficit, so the relevant test is at the most extreme drought actually
 observed. The largest SPEI-12 deficit in the panel is ≈ -2.0 (minimum -1.98), and
 25% of the baseline transmission slope is $0.25 \times 0.585 \approx 0.146$ SSI per unit SPEI
 (0.585 is the untreated baseline of the intent-to-treat panel; the downstream-gauge panel
 baseline is 0.531, and both appear in Supplementary Table S2); the maximum absolute
 perturbation of the streamflow-drought index a 25% slope alteration can produce is
 therefore $0.146 \times 1.98 \approx 0.29$ SSI units, below the 0.5-unit width of a single severity
 category. A 25% slope change thus cannot move a basin across an operational drought-
 severity threshold even under the worst observed deficit, and is marginal for drought
 management. We stress that 25% of the transmission slope nonetheless corresponds
 to a non-trivial absolute water volume: translating the maximal margin perturbation
 (0.29 SSI units) through each treated downstream gauge’s flow-to-SSI sensitivity gives
 a median of roughly 58 hm^3 over a 12-month season (interquartile range, IQR: 16 to
 137 hm^3 across the 40 gauges, about 11% of mean annual flow; the linear sensitivity
 overstates the dry-tail volume, so these are upper-end figures; Supplementary Table
 S16). A volume of this order is genuinely material for local water management, so the
 margin is anchored to the operational drought-severity classification, not to volumetric
 negligibility, and the causal weight of the null rests on the within-basin placebo rather
 than on the equivalence verdict. We adopt $\pm 25\%$ as this deliberately strict, pre-specified

margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at $\pm 25\%$) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the equivalence interval are built from the *standard deviation of the permutation-null distribution*, not the analytic standard error, since cluster-robust SEs over-reject at ~ 21 clusters (archived analysis code). To confirm the null is informative rather than underpowered, we additionally run a positive control. For each $\beta \in \{0, -0.15, -0.30, -0.45\}$ we set $SSI_{it}^* = SSI_{it} + \beta \times \text{treat}_i \times \text{SPEI}_{it}^c$, an additive perturbation on the standardized scale that injects a known buffering slope of magnitude β into treated units while leaving control observations and the panel’s error structure untouched, then re-fit the full estimator and randomization inference on each perturbed panel (Supplementary Fig. S2 and Table S3). $\beta = 0$ reproduces the observed null exactly, recovery is one-to-one in β , and detection is required once $|\beta|$ exceeds the MDE, verifying that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To locate where any apparent buffering comes from, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects); (4) the design

estimator with the baseline SPEI-to-outcome slope allowed to vary by Köppen region. It is important that these are distinct controls. The entropy balancing exact-matches *basin size and elevation* within Köppen main group; it does not balance the residual aridity gradient (which varies within B and C) nor impose a region-specific dose-response. The apparent buffering can therefore survive rungs (1) to (3), whose weighting leaves the within-region aridity gradient intact, and die only at rung (4), which lets each climate region have its own baseline transmission and so removes the between-region contrast that the matching does not. The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated downstream reaches against physically unregulated upstream reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo’s comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission

slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed. The linearity of that sensitivity is checked across the ~ 2000 m snowline with a piecewise fit among controls (no detectable break, interaction $p = 0.25$; Supplementary Table S20), and the snow regime itself is tested directly on the placebo estimand (Supplementary Table S12). To exclude that the pooled 2005–2024 slope averages an early-phase buffering (storage intact) into a late-phase failure (storage depleted), the buffering and placebo models are re-fit separately on the early (2010–2014) and late (2015–2024) megadrought phases, and the early-to-late change is tested as a $\text{treat} \times \text{SPEI} \times \text{phase}$ interaction on the pooled megadrought panel (year fixed effects absorb the phase level; Supplementary Table S24).

Inference

With only ~ 21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is *randomization inference*: the treatment label is permuted among units within Köppen $\times$ aridity- tercile strata (999 permutations) and the observed statistic is compared against the resulting null distribution; cluster-robust confidence intervals are reported alongside for completeness. Because hydrological and land-use outcomes can be spatially autocorrelated, we assess residual spatial dependence for each outcome with Moran’s I (great-circle distances, five-nearest-neighbour row-standardized weights, tested by 999 residual permutations). Where it is present, notably the water-rights

outcomes ($I \approx 0.4$ to 0.6 , $p = 0.001$; Supplementary Table S9), the unit-level stratified permutation can be anti-conservative because it does not preserve spatial structure. We therefore re-run the affected outcomes with a spatially-restricted permutation that swaps treatment at the watershed-block level, so co-located sub-watershed move together, and with watershed-block-clustered confidence intervals; this spatially-valid inference governs where the naive and spatial tests disagree (Supplementary Table S9). Parallel pre-trends are checked with a dynamic event study (dammed $\times$ year, reference 2009), and residual confounding is probed with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.

Software and reproducibility

All analyses are implemented in R within a **targets** pipeline (matched-set construction, estimators, and figures are individual targets), using **WeightIt**/**cobalt** for balancing and diagnostics, **fixest** for fixed-effects regression, **sf**/**terra**/**exactextractr** for the spatial extractions, and **renv** for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand. The complete R session information and package versions are archived alongside the pipeline output and reproducible via **renv::restore()**.

Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.21052149). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (SPEI-12 forcing and baseline aridity), MODIS MOD16 (evapotranspiration), MapBiomass Chile (land cover), the CR2 network (streamflow), ERA5-Land (snow water equivalent), SRTM (elevation), and the DGA reservoir, watershed, water-rights, and well-hydrograph registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

Alvarez-Garreton, Camila, Juan Pablo Boisier, René Garreaud, Jan Seibert, and Marc Vis. 2021. “Progressive Water Deficits During Multiyear Droughts in Basins with Long Hydrological Memory in Chile.” *Hydrology and Earth System Sciences* 25 (1): 429–46. <https://doi.org/10.5194/hess-25-429-2021>.

Ayala, Álvaro, Eduardo Muñoz-Castro, Daniel Farinotti, David Farías-Barahona, Pablo

712 A. Mendoza, Shelley MacDonell, James McPhee, Ximena Vargas, and Francesca
 713 Pellicciotti. 2025. “Less Water from Glaciers During Future Megadroughts in the
 714 Southern Andes.” *Communications Earth & Environment*. [https://doi.org/10.1038/](https://doi.org/10.1038/s43247-025-02845-6)
 715 [s43247-025-02845-6](https://doi.org/10.1038/s43247-025-02845-6).

716 Barriá, Pilar, Ignacio Barriá Sandoval, Carlos Guzman, Cristián Chadwick, Camila
 717 Alvarez-Garreton, Raúl Díaz-Vasconcellos, Anahí Ocampo-Melgar, and Rodrigo
 718 Fuster. 2021. “Water Allocation Under Climate Change.” *Elementa: Science of the*
 719 *Anthropocene* 9 (1): 00131. <https://doi.org/10.1525/elementa.2020.00131>.

720 Boisier, Juan P., Camila Alvarez-Garreton, Raúl R. Cordero, Alessandro Damiani,
 721 Laura Gallardo, René D. Garreaud, Fabrice Lambert, Cinthya Ramallo, Maisa Rojas,
 722 and Roberto Rondanelli. 2018. “Anthropogenic Drying in Central-Southern Chile
 723 Evidenced by Long-Term Observations and Climate Model Simulations.” *Elementa* 6
 724 (1): 74. <https://doi.org/10.1525/elementa.328>.

725 Di Baldassarre, Giuliano, Niko Wanders, Amir AghaKouchak, Linda Kuil, Sally Range-
 726 croft, Ted I. E. Veldkamp, Margaret Garcia, Pieter R. Van Oel, Korbinian Breinl,
 727 and Anne F. Van Loon. 2018. “Water Shortages Worsened by Reservoir Effects.”
 728 *Nature Sustainability* 1 (11): 617–22. <https://doi.org/10.1038/s41893-018-0159-0>.

729 Garreaud, René D., Camila Alvarez-Garreton, Jonathan Barichivich, Juan Pablo Boisier,
 730 Duncan Christie, Mauricio Galleguillos, Carlos LeQuesne, James McPhee, and Mauri-
 731 cio Zambrano-Bigiarini. 2017. “The 2010–2015 Megadrought in Central Chile:
 732 Impacts on Regional Hydroclimate and Vegetation.” *Hydrology and Earth System*

733 *Sciences* 21 (12): 6307–27. <https://doi.org/10.5194/hess-21-6307-2017>.

734 Garreaud, René, Juan Pablo Boisier, Camila Álvarez-Garretón, Duncan Christie, Tomás

735 Carrasco-Escaff, Iván Vergara, Roberto O. Chávez, et al. 2025. “Hyperdroughts in

736 Central Chile: Drivers, Impacts and Projections.” *Hydrometeorology/Instruments;*

737 *observation techniques*. <https://doi.org/10.5194/egusphere-2025-517>.

738 Götte, Jonas, and Manuela I. Brunner. 2024. “Hydrological Drought-To-Flood Transi-

739 tions Across Different Hydroclimates in the United States.” *Water Resources Research*

740 60 (7): e2023WR036504. <https://doi.org/10.1029/2023WR036504>.

741 Grafton, R. Q., J. Williams, C. J. Perry, F. Molle, C. Ringler, P. Steduto, B. Udall,

742 et al. 2018. “The Paradox of Irrigation Efficiency.” *Science* 361 (6404): 748–50.

743 <https://doi.org/10.1126/science.aat9314>.

744 Ho, Michelle, Upmanu Lall, Maura Allaire, Naresh Devineni, Hyun Han Kwon, Indrani

745 Pal, David Raff, and David Wegner. 2017. “The Future Role of Dams in the

746 United States of America.” *Water Resources Research* 53 (2): 982–98. [https:](https://doi.org/10.1002/2016WR019905)

747 [//doi.org/10.1002/2016WR019905](https://doi.org/10.1002/2016WR019905).

748 Kellner, Elke. 2021. “The Controversial Debate on the Role of Water Reservoirs in

749 Reducing Water Scarcity.” *WIREs Water* 8 (3): e1514. [https://doi.org/10.1002/](https://doi.org/10.1002/wat2.1514)

750 [wat2.1514](https://doi.org/10.1002/wat2.1514).

751 Kim, Kyung Y., Harihar Rajaram, and Venkataraman Lakshmi. 2025. “Observing

752 Decreasing Snow Cover and Increasing Surface Temperature Across the Andes with

753 Remotely Sensed and Reanalysis Data.” *Environmental Research Communications* 7:

754 021009. <https://doi.org/10.1088/2515-7620/adb382>.

755 Lee, Chaelim, Sinuk Kang, and Sangdan Kim. 2026. “Drought Resistance of Dams
756 Based on Propagation Analysis: A Case Study of Various Multipurpose Dams
757 in South Korea.” *Journal of Hydrology: Regional Studies* 63 (February): 103106.
758 <https://doi.org/10.1016/j.ejrh.2026.103106>.

759 Lema, Fabián, Pablo A. Mendoza, Nicolás A. Vásquez, Naoki Mizukami, Mauricio
760 Zambrano-Bigiarini, and Ximena Vargas. 2025. “Technical Note: What Does
761 the Standardized Streamflow Index Actually Reflect? Insights and Implications
762 for Hydrological Drought Analysis.” *Hydrology and Earth System Sciences* 29 (8):
763 1981–2002. <https://doi.org/10.5194/hess-29-1981-2025>.

764 Li, Yao, Gang Zhao, George H. Allen, and Huilin Gao. 2023. “Diminishing Storage
765 Returns of Reservoir Construction.” *Nature Communications* 14 (1): 3203. <https://doi.org/10.1038/s41467-023-38843-5>.

766

767 López-Moreno, J. I., S. M. Vicente-Serrano, S. Beguería, J. M. García-Ruiz, M. M. Portela,
768 and A. B. Almeida. 2009. “Dam Effects on Droughts Magnitude and Duration in a
769 Transboundary Basin: The Lower River Tagus, Spain and Portugal.” *Water Resources*
770 *Research* 45 (2): 2008WR007198. <https://doi.org/10.1029/2008WR007198>.

771 Macpherson, Elizabeth, Cristy Clark, Pía Weber Salazar, Natalie Baird, Afshin Akhtar-
772 Khavari, and Edward Challies. 2023. “Evolving Rights to (and of) Water in Chile: A
773 Case for Relationship-Based Water Law and Governance.” *The International Journal*
774 *of Human Rights*, October, 1–26. <https://doi.org/10.1080/13642987.2023.2266719>.

775 McKee, Thomas B., Nolan J. Doesken, and John Kleist. 1993. “The Relationship
776 of Drought Frequency and Duration to Time Scales.” *Proceedings of the Ninth*
777 *Conference on Applied Climatology, American Meteorological Society* 17 (22): 179–84.

778 Meza, Francisco J. 2013. “Recent Trends and {ENSO} Influence on Droughts in {N}or-
779 thern {C}hile: {A}n Application of the {S}tandardized {P}recipitation {E}vapotran-
780 spiration {I}ndex.” *Weather and Climate Extremes* 1: 51–58.

781 Oertel, Melanie, Francisco Javier Meza, and Jorge Gironás. 2020. “Observed Trends
782 and Relationships Between ENSO and Standardized Hydrometeorological Drought
783 Indices in Central Chile.” *Hydrological Processes* 34 (2): 159–74. [https://doi.org/10.](https://doi.org/10.1002/hyp.13596)
784 [1002/hyp.13596](https://doi.org/10.1002/hyp.13596).

785 Reade Malagueño, Benji, and Paolo D’Odorico. 2024. “¿Libre de La Maleza Estadista?
786 Assessing Neoliberal Promises and Water Markets in Chile.” *Earth’s Future* 12 (10):
787 e2024EF004963. <https://doi.org/10.1029/2024EF004963>.

788 Rivera, Diego, Alex Godoy-Faúndez, Mario Lillo, Amaya Alvez, Verónica Delgado,
789 Consuelo Gonzalo-Martín, Ernestina Menasalvas, Roberto Costumero, and Ángel
790 García-Pedrero. 2016. “Legal Disputes as a Proxy for Regional Conflicts over Water
791 Rights in Chile.” *Journal of Hydrology* 535 (April): 36–45. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.jhydrol.2016.01.057)
792 [jhydrol.2016.01.057](https://doi.org/10.1016/j.jhydrol.2016.01.057).

793 Rivera-Vidal, Rayen, José Luis Arumí, Ovidio Melo, Verónica Delgado, Víctor Parra, Ale-
794 jandra Stehr, and Linda Daniele. 2025. “Managed Aquifer Recharge Implementation
795 Challenges: Lessons from Chile’s Water-Scarce Regions.” *Groundwater for Sustainable*

796 *Development* 31 (November): 101502. <https://doi.org/10.1016/j.gsd.2025.101502>.

797 Scanlon, Bridget R., Sarah Fakhreddine, Ashraf Rateb, Inge De Graaf, Jay Famiglietti,
798 Tom Gleeson, R. Quentin Grafton, et al. 2023. “Global Water Resources and the Role
799 of Groundwater in a Resilient Water Future.” *Nature Reviews Earth & Environment*
800 4 (2): 87–101. <https://doi.org/10.1038/s43017-022-00378-6>.

801 Schilstra, Marco, Wen Wang, Pieter Richard Van Oel, Jingshu Wang, and Hui Cheng.
802 2024. “The Effects of Reservoir Storage and Water Use on the Upstream–Downstream
803 Drought Propagation.” *Journal of Hydrology* 631 (March): 130668. [https://doi.org/](https://doi.org/10.1016/j.jhydrol.2024.130668)
804 [10.1016/j.jhydrol.2024.130668](https://doi.org/10.1016/j.jhydrol.2024.130668).

805 Taucare, Matías, Benoît Viguié, Ronny Figueroa, and Linda Daniele. 2024. “The
806 Alarming State of Central Chile’s Groundwater Resources: A Paradigmatic Case of a
807 Lasting Overexploitation.” *Science of The Total Environment* 906 (January): 167723.
808 <https://doi.org/10.1016/j.scitotenv.2023.167723>.

809 Van Loon, Anne F. 2015. “Hydrological Drought Explained.” *WIREs Water* 2 (4):
810 359–92. <https://doi.org/10.1002/wat2.1085>.

811 Vicente-Serrano, Sergio M., Santiago Beguería, and Juan I. López-Moreno. 2010.
812 “A Multiscalar Drought Index Sensitive to Global Warming: The Standardized
813 Precipitation Evapotranspiration Index.” *Journal of Climate* 23 (7): 1696–1718.
814 <https://doi.org/10.1175/2009JCLI2909.1>.

815 Vicente-Serrano, Sergio M, Juan I López-Moreno, Santiago Beguería, Jorge Lorenzo-
816 Lacruz, Cesar Azorin-Molina, and Enrique Morán-Tejeda. 2012. “Accurate Com-

putation of a Streamflow Drought Index.” *J. Hydrol. Eng.* 17(2): 318–32. <http://hdl.handle.net/10261/49224>.

Vicente-Serrano, Sergio M., Tim R. McVicar, Diego G. Miralles, Yuting Yang, and Miquel Tomas-Burguera. 2019. “Unraveling the Influence of Atmospheric Evaporative Demand on Drought and Its Response to Climate Change.” *WIREs Climate Change*, December. <https://doi.org/10.1002/wcc.632>.

Wang, Wen, Maurits W. Ertsen, Mark D. Svoboda, and Mohsin Hafeez. 2016. “Propagation of Drought: From Meteorological Drought to Agricultural and Hydrological Drought.” *Advances in Meteorology*. <https://doi.org/10.1155/2016/6547209>.

Wang, Zhiwei, Liguang Jiang, Karina Nielsen, and Lei Wang. 2024. “Reservoir Filling Up Problems in a Changing Climate: Insights From CryoSat-2 Altimetry.” *Geophysical Research Letters* 51 (10): e2024GL108934. <https://doi.org/10.1029/2024GL108934>.

Webb, Mariana J., Jonathan M. Winter, Stephanie A. Spera, Jonathan W. Chipman, and Erich C. Osterberg. 2021. “Water, Agriculture, and Climate Dynamics in Central Chile’s Aconcagua River Basin.” *Physical Geography* 42 (5): 395–415. <https://doi.org/10.1080/02723646.2020.1790719>.

Wu, Jiefeng, Xingwei Chen, Huaxia Yao, Lu Gao, Ying Chen, and Meibing Liu. 2017. “Non-Linear Relationship of Hydrological Drought Responding to Meteorological Drought and Impact of a Large Reservoir.” *Journal of Hydrology* 551 (August): 495–507. <https://doi.org/10.1016/j.jhydrol.2017.06.029>.

Wu, Jiefeng, Zhiyong Liu, Huaxia Yao, Xiaohong Chen, Xingwei Chen, Yanhui Zheng,

838 and Yanhu He. 2018. “Impacts of Reservoir Operations on Multi-Scale Correlations
839 Between Hydrological Drought and Meteorological Drought.” *Journal of Hydrology*
840 563 (August): 726–36. <https://doi.org/10.1016/j.jhydrol.2018.06.053>.

841 Zambrano, Francisco, Anton Vrieling, Francisco Meza, Iongel Duran-Llacer, Francisco
842 Fernández, Alejandro Venegas-González, Nicolas Raab, and Dylan Craven. 2025.
843 “From Drought to Aridification: Land-Cover Fingerprints of a Drying Chile.” *Earth’s*
844 *Future* 13 (12): e2025EF006744. <https://doi.org/10.1029/2025EF006744>.

845 Zambrano, Francisco, Anton Vrieling, Andy Nelson, Michele Meroni, and Tsegaye Tadesse.
846 2018. “Prediction of Drought-Induced Reduction of Agricultural Productivity in Chile
847 from MODIS, Rainfall Estimates, and Climate Oscillation Indices.” *Remote Sensing*
848 *of Environment* 219 (December): 15–30. <https://doi.org/10.1016/j.rse.2018.10.006>.

849 Zambrano, Francisco, Brian Wardlow, Tsegaye Tadesse, Mario Lillo-Saavedra, and Oc-
850 tavio Lagos. 2017. “Evaluating Satellite-Derived Long-Term Historical Precipitation
851 Datasets for Drought Monitoring in Chile.” *Atmospheric Research* 186 (April): 26–42.
852 <https://doi.org/10.1016/j.atmosres.2016.11.006>.